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✓ Day 21: Early Stopping in TensorFlow

1. Introduction to Early Stopping

Early Stopping is a form of regularization that halts the training process when the validation performance stops improving. It's essential for preventing overfitting and saving computation time.

✓ 2. Building a Neural Network with Early Stopping

```
import tensorflow as tf
import numpy as np

# Generating a simple dataset
X_train = np.random.rand(200, 5) # 200 samples, 5 features
y_train = np.random.rand(200, 1) # 200 target values

# Creating an EarlyStopping callback
early_stopping = tf.keras.callbacks.EarlyStopping(
    monitor='val_loss', # Monitors validation loss
    patience=3, # Stops training after 3 epochs of no improvement
    restore_best_weights=True # Restores weights from the epoch with the best validation loss
)

# Building a simple neural network model
model = tf.keras.Sequential([
    tf.keras.layers.Dense(64, activation='relu', input_shape=(5,)),
    tf.keras.layers.Dense(32, activation='relu'),
    tf.keras.layers.Dense(1)
])

# Compiling the model
model.compile(optimizer='adam', loss='mean_squared_error')

# Training the model with Early Stopping
```

```
# Training the model with Early Stopping
```

```
history = model.fit(X_train, y_train, epochs=50, batch_size=32, validation_split=0.2, callbacks=[early_stopping])
```

```
↳ /usr/local/lib/python3.10/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an `input_shape`/`input_dim` argument to `super().__init__(activity_regularizer=activity_regularizer, **kwargs)`
```

```
Epoch 1/50
```

```
5/5 ————— 7s 86ms/step - loss: 0.4650 - val_loss: 0.3024
```

```
Epoch 2/50
```

```
5/5 ————— 0s 16ms/step - loss: 0.3396 - val_loss: 0.2150
```

```
Epoch 3/50
```

```
5/5 ————— 0s 12ms/step - loss: 0.2492 - val_loss: 0.1499
```

```
Epoch 4/50
```

```
5/5 ————— 0s 13ms/step - loss: 0.1587 - val_loss: 0.1049
```

```
Epoch 5/50
```

```
5/5 ————— 0s 11ms/step - loss: 0.1142 - val_loss: 0.0809
```

```
Epoch 6/50
```

```
5/5 ————— 0s 11ms/step - loss: 0.1011 - val_loss: 0.0778
```

```
Epoch 7/50
```

```
5/5 ————— 0s 17ms/step - loss: 0.0810 - val_loss: 0.0844
```

```
Epoch 8/50
```

```
5/5 ————— 0s 11ms/step - loss: 0.0968 - val_loss: 0.0866
```

```
Epoch 9/50
```

```
5/5 ————— 0s 15ms/step - loss: 0.1021 - val_loss: 0.0816
```

3. Evaluating the Model Performance

```
# Generating test data
```

```
X_test = np.random.rand(50, 5)
```

```
y_test = np.random.rand(50, 1)
```

```
# Evaluating the model on test data
```

```
test_loss = model.evaluate(X_test, y_test)
```

```
print(f"Test Loss with Early Stopping: {test_loss}")
```

```
↳ 2/2 ————— 0s 6ms/step - loss: 0.0823
```

```
Test Loss with Early Stopping: 0.08304407447576523
```

```
import matplotlib.pyplot as plt
```

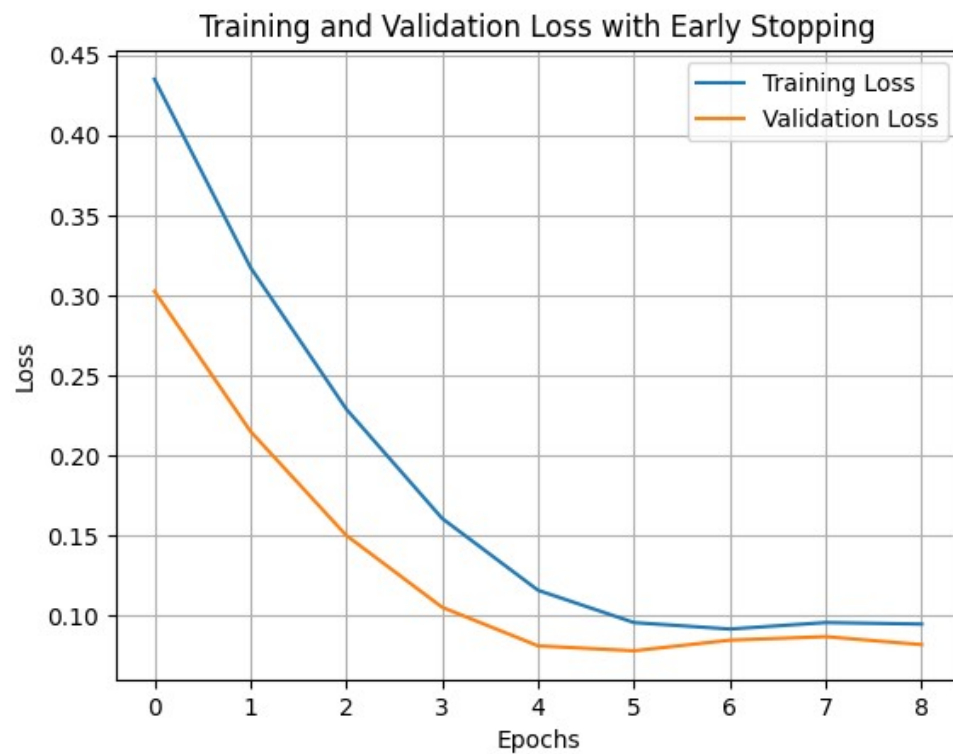
```
# Plotting the loss curves
```

```
plt.plot(history.history['loss'], label='Training Loss')
```

```
plt.plot(history.history['val_loss'], label='Validation Loss')
```

```
# plt.show() # To display the plot (Note: This line is commented out in the original image)
```

```
plt.title('Training and Validation Loss with Early Stopping')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)
plt.show()
```



Key Benefits of Early Stopping:

1. Prevents overfitting by halting training when the model starts to overtrain on the data.
2. Reduces training time by avoiding unnecessary epochs.
3. Restores the model to the best weights found during training.

✓ Conclusion

Early Stopping is a simple yet powerful technique that enhances your model's performance while making the training process more efficient. By stopping training at the right time, you avoid overfitting and save valuable time and resources.

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